SLEEP EFFICIENCY – A STUDY OF SLEEP EFFICIENCY AND SLEEP PATTERNS

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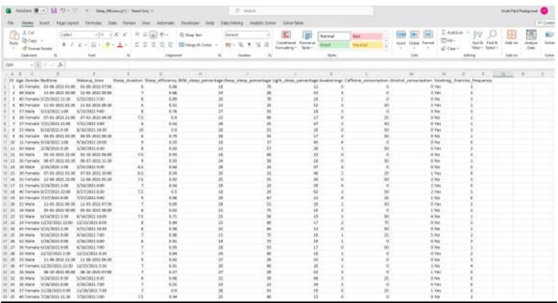
UNDER THE GUIDANCE OF **PROF. RICHARD SIGMAN**

INTRODUCTION:

Good sleep is essential in the fast-paced world of today. This research uses regression approaches to analyse the intricate correlations within sleep data to investigate sleep efficiency, which is a significant component representing the efficacy of the sleep cycle. Finding quantitative insights into the variables influencing sleep efficiency is the aim. Regression analysis allows us to go beyond descriptive data in our search for important variables that either enhance or diminish sleep efficiency. This data-driven method offers insights into potential therapies to promote sleep health and advances our understanding of sleep dynamics in addition to having potential practical implications. In a time where work can occasionally trump relaxation, this regression-focused investigation provides both empirical evidence and practical recommendations for those looking to enhance their sleep.

DATASET:

Figure 1:



SOURCE:

The popular platform Kaggle, which offers a large selection of datasets for group data science projects, is where we got the dataset, we utilized for our regression study. Many datasets donated by the international data science community are housed on Kaggle, allowing researchers and analysts to access a multitude of data in a variety of fields. For this study, we chose a particular dataset as it fit our research goals and had factors that support them.

DESCRIPTION:

The popular platform Kaggle, which offers a large selection of datasets for group data science projects, is where we got the dataset, we utilized for our regression study. Many datasets donated by the international data science community are housed on Kaggle, allowing researchers and analysts to access a multitude of data in a variety of fields. For this study, we chose a particular dataset as it fit our research goals and had factors that support them. Being an open data platform, Kaggle encourages openness and reproducibility among scientists while also facilitating the investigation of novel research problems. By using Kaggle's resources, we can be sure that our research is robust because the dataset is reviewed and validated by other data aficionados, which adds to the validity of our conclusions.

CONTEXT:

Our dataset, which we obtained from Kaggle, is an invaluable information resource for researching sleep efficiency. By including data about sleep length, patterns, and possible affecting factors, it enables us to explore the intricate dynamics of sleep quality. Our approach is predicated on the expanding understanding of how critical sleep is to contemporary lifestyles. People's lives are moving at an ever-increasing pace, so it's critical to understand what influences how well they sleep. The information was meticulously selected to guarantee completeness and relevance, which put us in a good position to find patterns and correlations that provide insight into the best ways to sleep. Our research focuses on elements like lifestyle decisions, environmental influences, and bedtime practices to offer practical advice to those looking to enhance their quality of sleep in the face of contemporary difficulties. This setting is consistent with the larger conversation about health and wellness, which highlights the critical role sleep plays in sustaining a resilient and well-balanced way of life.

RESEARCH QUESTIONS:

1. Do sleep patterns and efficiency differ significantly between gender groups in the dataset?

2. How does caffeine consumption relate to sleep efficiency and the number of awakenings in the dataset?
3. Is there a significant association between exercise frequency and the percentage of REM sleep in the dataset?

DATA ANALYSIS

METHODS USED:

- TYPE-1 **LINEAR REGRESSION**.
- TYPE-2 **CLUSTERING.**

LINEAR REGRESSION:

Data Preprocessing: The dataset is read into R, and potential space strings are treated as missing values (NA). Empty values are filled with 0, and categorical variables, namely Gender and Smoking status, are converted to factors.

Linear Regression Basics: Start with a simple linear regression using the 'lm' function, examining the relationship between Sleep Efficiency and Age. A summary of the regression model is then displayed.

Multiple Linear Regression: Extended the analysis by performing linear regression for Sleep Efficiency against each numerical variable in the dataset. This is achieved through a loop that iterates over each numeric column, excluding the 'ID' and 'Sleep_efficiency' columns.

Visualizing Regression: For a more intuitive understanding, we create scatter plots with regression lines for each numeric variable against Sleep Efficiency. The plots provide a visual representation of the relationship between the variables.

Research Questions & work done:

Question 1: Gender and Sleep Efficiency

I have explored the impact of gender on sleep efficiency through linear regression. The 'lm' function is used to create a model, and the 'ggplot2' library is employed to generate a scatter plot with a regression line.

Question 2: Caffeine Consumption and Sleep Patterns

The relationship between caffeine consumption, sleep efficiency, and the number of awakenings is examined using multiple linear regression. Two separate plots are generated to visualize the impact on both Sleep Efficiency and Awakenings.

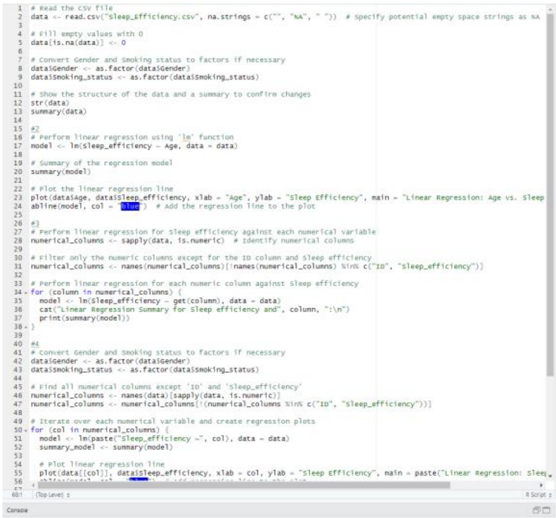
Question 3: Exercise Frequency and REM Sleep

I have investigated the association between exercise frequency and the percentage of REM sleep. A linear regression model is built, and a corresponding scatter plot illustrates the relationship.

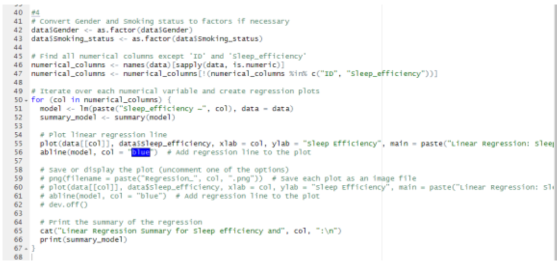
Conclusion: Through this analysis, we aim to gain insights into the factors influencing sleep efficiency. The combination of descriptive statistics, linear regression models, and visualizations provides a comprehensive understanding of the dataset and addresses specific research questions.

Data pre-processing, basic linear regression, and multiple linear regression:

Figure1(a):



```r
# Read the csv file
data <- read.csv("Sleep_Efficiency.csv", na.strings = c("", "NA", " "))  # Specify potential empty space strings as NA

# Fill empty values with 0
data[is.na(data)] <- 0

# Convert Gender and Smoking status to factors if necessary
data$Gender <- as.factor(data$Gender)
data$Smoking_status <- as.factor(data$Smoking_status)

# Show the structure of the data and a summary to confirm changes
str(data)
summary(data)

#2
# Perform linear regression using 'lm' function
model <- lm(Sleep_efficiency ~ Age, data = data)

# Summary of the regression model
summary(model)

# Plot the linear regression line
plot(data$Age, data$Sleep_efficiency, xlab = "Age", ylab = "Sleep Efficiency", main = "Linear Regression: Age vs. Sleep
abline(model, col = "blue")  # Add the regression line to the plot

#3
# Perform linear regression for Sleep efficiency against each numerical variable
numerical_columns <- sapply(data, is.numeric)  # Identify numerical columns

# Filter only the numeric columns except for the ID column and Sleep efficiency
numerical_columns <- names(numerical_columns)[!names(numerical_columns) %in% c("ID", "Sleep_efficiency")]

# Perform linear regression for each numeric column against Sleep efficiency
for (column in numerical_columns) {
  model <- lm(Sleep_efficiency ~ get(column), data = data)
  cat("Linear Regression Summary for Sleep efficiency and", column, ":\n")
  print(summary(model))
}

#4
# Convert Gender and Smoking status to factors if necessary
data$Gender <- as.factor(data$Gender)
data$Smoking_status <- as.factor(data$Smoking_status)

# Find all numerical columns except 'ID' and 'Sleep_efficiency'
numerical_columns <- names(data)[sapply(data, is.numeric)]
numerical_columns <- numerical_columns[!(numerical_columns %in% c("ID", "Sleep_efficiency"))]

# Iterate over each numerical variable and create regression plots
for (col in numerical_columns) {
  model <- lm(paste("Sleep_efficiency ~", col), data = data)
  summary_model <- summary(model)

  # Plot linear regression line
  plot(data[[col]], data$Sleep_efficiency, xlab = col, ylab = "Sleep Efficiency", main = paste("Linear Regression: Slee
```



```r
#4
# Convert Gender and Smoking status to factors if necessary
data$Gender <- as.factor(data$Gender)
data$Smoking_status <- as.factor(data$Smoking_status)

# Find all numerical columns except 'ID' and 'Sleep_efficiency'
numerical_columns <- names(data)[sapply(data, is.numeric)]
numerical_columns <- numerical_columns[!(numerical_columns %in% c("ID", "Sleep_efficiency"))]

# Iterate over each numerical variable and create regression plots
for (col in numerical_columns) {
  model <- lm(paste("Sleep_efficiency ~", col), data = data)
  summary_model <- summary(model)

  # Plot linear regression line
  plot(data[[col]], data$Sleep_efficiency, xlab = col, ylab = "Sleep Efficiency", main = paste("Linear Regression: Slee
  abline(model, col = "blue")  # Add regression line to the plot

  # Save or display the plot (uncomment one of the options)
  # png(filename = paste("Regression_", col, ".png"))  # Save each plot as an image file
  # plot(data[[col]], data$Sleep_efficiency, xlab = col, ylab = "Sleep Efficiency", main = paste("Linear Regression: Sl
  # abline(model, col = "blue")  # Add regression line to the plot
  # dev.off()

  # Print the summary of the regression
  cat("Linear Regression Summary for Sleep efficiency and", col, ":\n")
  print(summary_model)
}
```

OUTPUTS:

Figure 1.1:

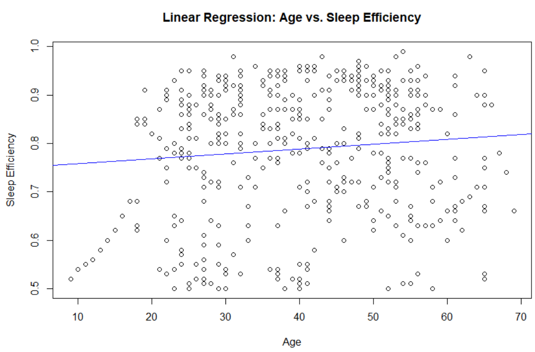


Figure 1.1 indicates that Age and the effectiveness of sleep are positively correlated. This indicates that as people age, their ability to sleep gets better on average. The average sleep efficiency rises by around 0.025 units for every year of age, according to the linear regression line. The dispersion of the blue data points around the red regression line indicates a high degree of variance in the data. This indicates that a variety of variables besides age affect how well people sleep.

<u>Some other basic plots:</u>

<u>Figure 2.1:</u>

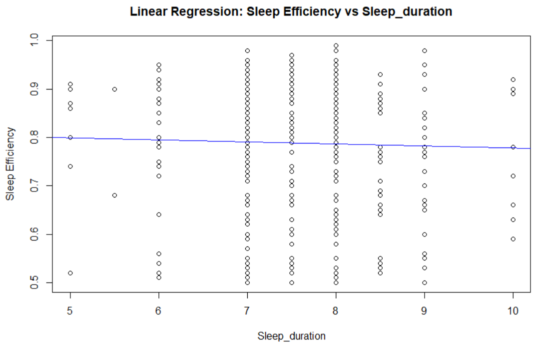


Figure 2.1 represents that the relation has an intercept of 0.8 and is negatively co-related, as the sleep duration increases the sleep efficiency decreases but there is no steep decrease.

<u>Figure 2.2:</u>

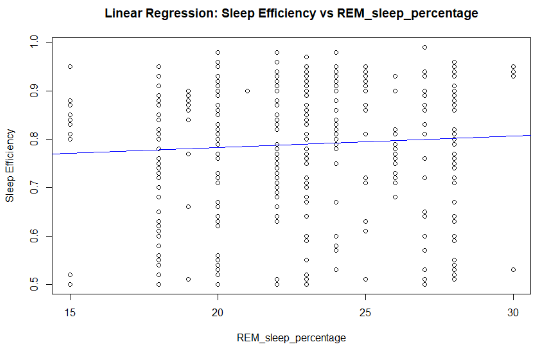


Figure 2.2 shows that the correlation between sleep efficiency and REM sleep percentage has a positive relationship.

<u>Figure 2.3:</u>

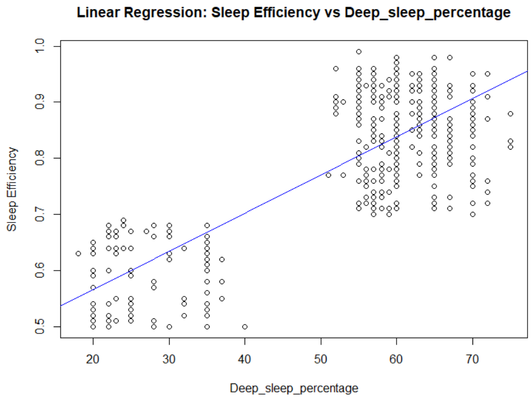


Figure 2.3 shows that sleep efficiency and deep sleep percentage have a strong positive correlation.

<u>Figure 2.4:</u>

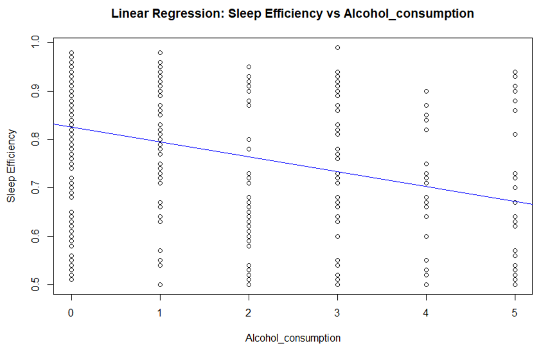


Figure 2.4 indicates that the graph has a decreasing slope of sleep efficiency concerning alcohol consumption.

<u>Figure 2.5:</u>

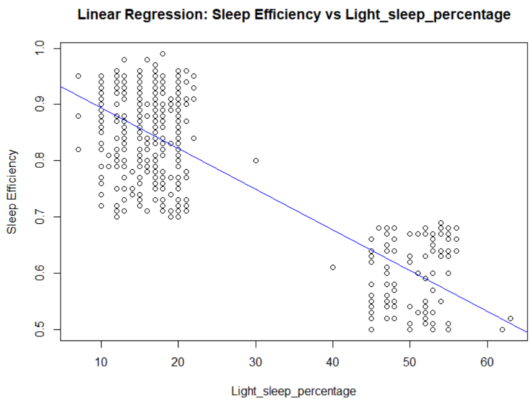


Figure 2.5 shows that there is a negative correlation.

<u>NOTE:</u>

I have also done a few other basic plots but I felt these plots gave me accurate results so I have not considered those plots.

LINEAR REGRESSION BASED ON RESEARCH QUESTIONS (2,3):

Figure 3:

```
# Convert Gender and Smoking status to factors if necessary
data$Gender <- as.factor(data$Gender)
data$Smoking_status <- as.factor(data$Smoking_status)

# Question 2: How does caffeine consumption relate to sleep efficiency and the number of awakenings?
model_caffeine <- lm(Sleep_efficiency ~ Caffeine_consumption + Awakenings, data = data)

# Summary of the regression model
summary(model_caffeine)

# Plot the linear regression line for Caffeine Consumption and Sleep Efficiency
plot(data$Caffeine_consumption, data$Sleep_efficiency, xlab = "Caffeine Consumption", ylab = "Sleep Efficiency", main = "Linear Regression: Caffeine Consumption vs. Sleep Efficiency")
abline(model_caffeine, col = "blue")  # Add the regression line to the plot

# Plot the linear regression line for Caffeine Consumption and Awakenings
plot(data$Caffeine_consumption, data$Awakenings, xlab = "Caffeine Consumption", ylab = "Awakenings", main = "Linear Regression: Caffeine Consumption vs. Awakenings")
abline(model_caffeine, col = "blue")  # Add the regression line to the plot

# Question 3: Is there a significant association between exercise frequency and the percentage of REM sleep?
model_exercise_rem <- lm(REM_sleep_percentage ~ Exercise_frequency, data = data)

# Summary of the regression model
summary(model_exercise_rem)

# Plot the linear regression line for Exercise Frequency and REM Sleep Percentage
plot(data$Exercise_frequency, data$REM_sleep_percentage, xlab = "Exercise Frequency", ylab = "REM Sleep Percentage", main = "Linear Regression: Exercise Frequency vs. REM Sleep Percentage")
abline(model_exercise_rem, col = "blue")  # Add the regression line to the plot
```

I have investigated the association between exercise frequency and the percentage of REM sleep. A linear regression model is built, and a corresponding scatter plot illustrates the relationship. The relationship between caffeine consumption, sleep efficiency, and the number of awakenings is examined using multiple linear regression. Two separate plots are generated to visualize the impact on both Sleep Efficiency and Awakenings.

RESEARCH QUESTION PLOTS (2,3):

Figure 3.1:

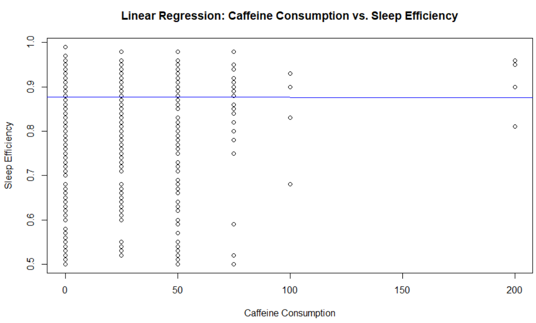
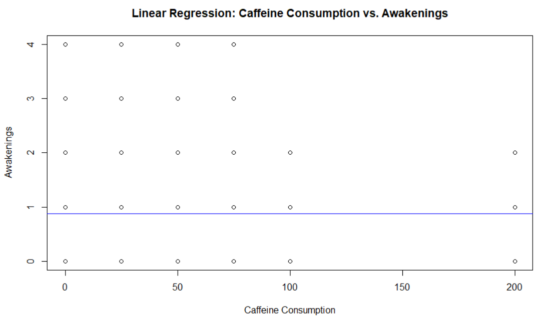


Figure 3.2:



Figures 3.1,3.2 show that there is no significance of caffeine consumption concerning awakening and sleep efficiency as both remain constant even when caffeine consumption increases.

Figure 3.3:

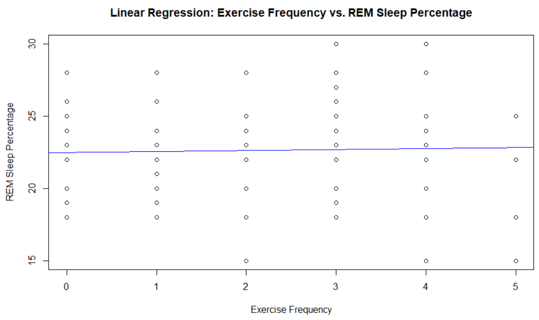


Figure 3.3 shows that there is a very slight increase in the slope of REM sleep percentage w.r.t exercise frequency.

CLUSTERING:

Introduction:

In this section, we explore clustering techniques to identify patterns within the "Sleep_Efficiency" dataset. Unlike linear regression, which focuses on relationships between variables, clustering aims to group similar data points. The clustering analysis is performed individually without group involvement.

K-Means Clustering by Gender:

Libraries and Data Loading: The necessary libraries, including 'cluster,' are loaded. The dataset, "Sleep_Efficiency.csv," is read into R.

Data Selection and Standardization: I chose relevant variables for clustering, such as Sleep Duration and REM Sleep Percentage. The data is then standardized to ensure that all variables contribute equally to the clustering process.

K-Means Clustering

K-means clustering is performed based on Gender, assuming Gender is a binary variable (centers = 2). The resulting clusters are visualized by plotting the standardized data points in different colors corresponding to their assigned clusters. Cluster centers are also marked on the plot.

Conclusion: In conclusion, clustering is a valuable tool for exploratory analysis, helping to identify inherent structures within datasets and facilitating a deeper understanding of the underlying patterns.

RESEARCH QUESTION PLOT (1):

Figure4:

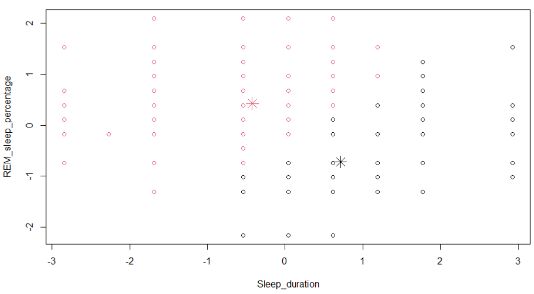


Figure4.1:

The two graphs are the same, but Figure 4 is not labelled, and Fig 4.1 is labelled, females have higher percentage values and males have higher sleep duration.

CHALLENGES:

The sleep efficiency analysis project faced significant challenges due to the subjective nature of sleep-related data, compounded by individual variations and potential biases. Linear regression, addressing Research Questions 2 and 3, navigated complex interactions between physiological and environmental factors, hindered by lifestyle choices like coffee consumption and screen time. Clustering, focusing on Research Question 1 regarding gender's impact on sleep efficiency, grappled with intricate individual sleep habits and potential biases. Meticulous examination of anomalies and missing data was crucial for research validity. Despite these complexities, overcoming challenges proved essential for advancing sleep health understanding and contributing valuable insights to the field in 100 words.

CONCLUSION:

To sum up, because sleep-related data is subjective, overcoming obstacles was crucial to the sleep efficiency study endeavour. Individual differences, biases in self-reporting, and complicated interplay between physiological and environmental variables added to the complexity. It was difficult to isolate certain factors like lifestyle choices (e.g., coffee use) using linear regression, which addressed Research Questions 2 and 3. Clustering struggled with complex sleep patterns and biases, with an emphasis on Research Question 1 about gender. Validity required a close inspection of anomalies and missing information. Overcoming obstacles improved our knowledge of sleep health despite its complexity and provided insightful information to the field.

References:

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2. H. Wickham. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York, 2 016.
3. Kaggle: Sleep Efficiency Dataset (kaggle.com)
4. R-Pubs: **https://rpubs.com/stanislausz/se2**
5. R Core Team (2023). _R: A Language and E nvironment for Statistical Computing_. R F oundation for Statistical Computing, Vienn a, Austria. https://www.R-project.org/
6. Maechler, M., Rousseeuw, P., Struyf, A., H ubert, M., Hornik, K.(2022). cluster: Clust er Analysis Basics and Extensions. R packa ge version 2.1.4.
7. I have used ChatGPT for paraphrasing and a few summaries. https://chat.openai.com/ OpenAI. (2023). ChatGPT. San Franciso: OpenAI.

I have used ChatGPT for paraphrasing and a few summaries.